

ORLANDO EXECUTIVE, FL, USA

WMO: 722053

Lat: 28.545N

Lon: 81.333W

Elev: 33

StdP: 100.93

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 12841

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 3.9	(c) 6.3	(d) -7.0	(e) 2.1	(f) 8.6	(g) -3.8	(h) 2.8	(i) 10.0	(j) 10.4	(k) 17.2	(l) 9.2	(m) 17.6	(n) 3.3	(o) 340	(p) 0.371

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 8.3	(c) 34.3	(d) 24.6	(e) 33.6	(f) 24.5	(g) 32.8	(h) 24.4	(i) 26.7	(j) 30.4	(k) 26.2	(l) 30.0	(m) 25.8	(n) 29.7	(o) 3.4	(p) 270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.9	21.3	27.9	25.2	20.4	27.5	24.9	20.0	27.4	82.8	30.1	81.3	30.0	79.7	29.9	29.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.8	(b) 7.9	(c) 7.1	(e) 0.4	(f) 35.9	(g) 2.2	(h) 1.1	(i) -1.2	(j) 36.6	(k) -2.5	(l) 37.3	(m) -3.8	(n) 37.9	(o) -5.4	(p) 38.7		
(a) 8.8	(b) 7.9	(c) 7.1	(e) -2.1	(f) 27.6	(g) 2.1	(h) 1.0	(i) -3.6	(j) 28.3	(k) -4.8	(l) 28.9	(m) -5.9	(n) 29.5	(o) -7.4	(p) 30.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	23.1	16.0	18.0	20.0	22.7	25.7	27.6	28.4	28.4	27.5	24.6	20.4	17.9		
	DBStd	5.30	4.56	4.28	3.69	2.87	2.15	1.59	1.21	1.16	1.29	2.80	3.45	4.43		
	HDD10.0	14	8	3	0	0	0	0	0	0	0	0	0	3		
	HDD18.3	271	98	54	27	5	1	0	0	0	0	0	20	64		
	CDD10.0	4805	196	225	310	381	486	528	571	571	524	453	312	248		
	CDD18.3	2020	27	43	79	136	228	278	313	313	274	197	82	50		
	CDH23.3	17408	133	256	523	1013	1971	2582	3175	3175	2498	1430	422	231		
(14) Wind	WSAvg	3.2	3.3	3.4	3.7	3.7	3.4	3.0	2.7	2.7	3.0	3.2	3.2	3.1		
	PrecAvg	1295	62	60	88	59	87	199	185	190	170	95	46	54		
	PrecMax	1735	148	223	287	183	433	439	328	457	356	260	200	248		
	PrecMin	845	3	11	0	2	11	43	69	81	84	12	1	10		
	PrecStd	232	41	45	73	36	80	79	67	84	70	58	44	61		
	DB	27.7	29.5	30.8	32.4	34.3	36.1	35.2	35.0	33.8	32.4	30.0	28.6			
	MCWB	20.7	21.0	21.5	21.7	22.6	24.6	25.2	25.1	24.5	24.9	22.9	22.1			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	26.3	27.9	29.1	31.1	32.9	34.1	34.0	34.0	32.8	31.3	28.5	27.3			
	2% MCWB	19.7	20.3	20.4	21.4	22.3	24.4	24.9	25.0	24.4	24.0	22.2	21.1			
	5% DB	24.9	26.6	27.8	29.8	32.2	33.0	33.2	33.0	32.2	30.2	27.3	26.1			
	5% MCWB	19.0	19.5	19.7	20.7	22.2	24.3	24.9	25.0	24.4	23.4	21.1	20.5			
	10% DB	23.0	24.8	26.5	28.4	31.0	32.0	32.4	32.3	31.2	28.9	26.0	24.4			
	10% MCWB	18.4	18.7	19.0	20.2	22.0	24.1	24.9	25.1	24.3	22.8	20.1	19.9			
	WB	22.3	22.4	23.4	24.0	25.2	26.6	27.2	27.2	26.9	26.4	24.4	23.3			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% MCDB	25.6	26.2	27.3	28.6	30.0	31.7	31.3	31.1	29.7	29.5	27.6	26.5			
	2% WB	21.0	21.6	22.2	23.0	24.4	25.8	26.4	26.6	26.3	25.5	23.5	22.4			
	2% MCDB	24.2	25.5	26.4	27.7	29.1	30.6	30.6	30.4	29.4	28.6	26.4	25.4			
	5% WB	20.2	20.8	21.3	22.4	23.9	25.5	26.0	26.2	25.8	24.9	22.6	21.6			
	5% MCDB	23.4	24.6	25.4	26.9	28.5	30.1	30.1	30.0	29.1	27.9	25.4	24.3			
	10% WB	19.3	20.0	20.5	21.7	23.3	25.1	25.6	25.8	25.4	24.3	21.7	20.8			
	10% MCDB	22.4	23.7	24.5	26.1	27.8	29.5	29.8	29.6	28.8	27.3	24.6	23.4			
(35) Mean Daily Temperature Range	MDBR	10.5	10.6	10.7	10.7	10.1	8.8	8.6	8.3	7.8	8.3	9.4	9.6			
	5% DB MCDBR	10.9	11.4	11.3	11.3	11.0	10.1	9.3	9.1	8.7	8.6	9.5	9.8			
	5% DB MCWBR	5.4	4.8	4.4	3.9	3.2	3.0	2.9	2.8	2.6	3.0	4.3	4.5			
	5% WB MCDBR	9.7	9.8	9.8	10.2	9.6	8.9	8.7	8.3	7.7	7.4	8.1	8.5			
	5% WB MCWBR	5.8	5.0	5.0	4.4	3.6	3.2	3.1	3.1	2.9	3.1	4.3	4.8			
(40) Clear-Sky Solar Irradiance	taub	0.339	0.352	0.373	0.391	0.425	0.455	0.476	0.465	0.421	0.396	0.357	0.346			
	taud	2.525	2.496	2.428	2.356	2.309	2.298	2.236	2.284	2.398	2.433	2.496	2.526			
	Ebn at Noon	904	919	916	905	870	839	820	828	857	859	874	877			
	Edh at Noon	90	100	113	125	132	133	141	133	115	105	92	86			
(44) All-Sky Solar Radiation	RadAvg	3.41	4.04	5.12	6.08	6.30	5.66	5.67	5.38	4.84	4.40	3.62	3.11			
	RadStd	0.25	0.27	0.39	0.31	0.42	0.53	0.33	0.36	0.28	0.31	0.19	0.24			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	-48	+178	N/A
(47) Regional (44 neighbors)		+0.43	N/A	N/A	N/A	N/A	N/A	-5	-47	+160	+116

Nomenclature: See separate page